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Risk Assessment Details ID: RA_SCElse_103574 Rev No: 3  Draft ID: 0 			
User's School/ Department/ Student Activity Group:	Singapore Centre for Environmental Life Sciences Engineering	Project Title:	Use of Super Resolution ELYRA & CLSM 780 Microscope
Workplace:	Singapore Centre for Environmental Life Sciences Engineering	Other workplace:	-
Location:	Advanced Biofilm Imaging Facility	Conducted By:	Radek Machan(RADEK.MACHAN);Foo Yong Hwee(YHFOO)
Approved By:	Peter Torok (PETER.TOROK@ntu.edu.sg)	Submitted By/Submitted Date:	Foo Yong Hwee/26-Feb-26
Approved Date:	03-Mar-26	Next Review Date:	02-Mar-29
Status:	Approved	Comments:	-

1. Hazard Identification					2. Risk Evaluation				3. Risk Control				
1a.	1b.	1c.	1d.	1e.	2a.	2b.	2c.	2d.	3a.	3b.	3c.	3d.	3e.
No.	Work Activity	Hazard	Sub Hazard	Possible Accident/III Health & Person-at-Risk	Existing Risk Control	S	L	R	Additional Risk Control	S	L	R	Follow Up by & date
1	Use of Super Resolution ELYRA & CLSM 780 Microscope	Mechanical	Cut	Others:Cuts, lacerations or puncture from sharp edges of coverslips and slides	Laboratory worker shall remain vigilant and handle coverslips and slides with care. Laboratory worker shall successfully complete ? Basic Safety Training? e-Learning module in the NTULearn portal or Workday@NTU. Wear the appropriate PPE; Lab coat, long pants, nitrile gloves & covered shoes. Additional PPE may include safety eyewear & thicker gloves. Dispose specimen in sharps bin immediately after use.	2	2	4	NA				
		Biological	Contact with or infection by bacteria, virus, fungal spores or toxin	Others:Depending on the properties of the chemicals or biological materials, contact may be cytotoxic to workers, cause irritations, infections or other injuries.	Slide preparation shall be conducted in the designated main laboratory and not in the imaging facility, which is meant to be a dry lab facility. Dispose specimen in the sharps bin immediately after use. Laboratory workers shall inform the person-in-charge of the imaging facility beforehand should any toxic substances or any Schedule 1 or 2 biological agents (not allowed in SCElse) are used; Laboratory workers are expected to submit a detailed risk assessment (with approval from supervisor and/or cluster head) of their activities involving the substance/agents;	3	1	3	NA				

			Laboratory worker shall successfully complete ? Basic Biosafety Training? e-Learning module in the NTULearn portal or Workday@NTU; and Wear the appropriate PPE; Lab coat, long pants, nitrile gloves & covered shoes. Additional PPE may include safety eyewear & thicker gloves. Wear mask whenever appropriate.				
Chemical	Others:Chemical hazard. Fluorescent dye used may be cytotoxic.	Others:Fluorescent dye used may cause skin irritation or may be mildly cytotoxic when in contact.	Staining procedures shall be conducted in the designated main laboratory and not in the imaging facility, which is meant to be a dry lab facility. Laboratory worker shall successfully complete ? Chemical Safety Training? e-Learning module in the NTULearn portal or Workday@NTU. Laboratory worker shall read the SDS of the fluorescent dye used and wear the appropriate PPE: Lab coat, long pants & covered shoes. Nitrile gloves must be used when handling the specimen. Additional PPE may include safety eyewear, thicker gloves, double-glove and mask, where appropriate.	3	1	3	NA
Physical	Non-ionisation radiation	Eye injury	The CLSM system comes with a Plexiglas reflection shield or a non-transparent enclosure to protect workers from stray laser beams. The CLSM system comes with a safety interlock feature to prevent worker exposure to direct laser beam; laser beam can never enter the microscope eyepiece. Appropriate warning signs are in place to warn workers not to look directly into the laser beam. All workers are required to hold a valid NEA N3 license, which ensures that they familiarise themselves with fundamentals of non-ionising radiation safety and undergone an eye checkup before being allowed to operate the microscope. Besides that, workers are required to take an online non-ionising radiation safety module (Working with Non-Ionising Radiation - OHS2NIR01) every 3 years.	3	1	3	NA
Ergonomic	Awkward posture	Sprains, strains	Maintain good posture at all times while working at the CLSM. Ensure that the chair is adjusted to a comfortable height. Take breaks in between the long sessions on the CLSM. Do not work at the microscope/computer for more than two hours without taking a break (SCELSE recommends workers to take a 10 mins break with every 50 mins of microscopy time). Laboratory worker shall successfully complete ?	3	1	3	NA

			Basic Safety Training? e-Learning module in the NTULearn portal or Workday@NTU.				
Electrical	Contact with electrical energy	Others:Electrocution	Conduct visual inspection of electrical plugs, wires & cables before use. Use approved & suitable electrical plugs with the `SAFETY MARK` logo. Workers shall ensure that the area surrounding the equipment is kept dry. Place cables in cable organizer where necessary to protect components from moisture. Laboratory worker shall successfully complete ? Basic Safety Training? e-Learning module in the NTULearn portal or Workday@NTU.	5	1	5	NA
Chemical	Sensitizer or irritant	Others:Skin irritation from Immersion oil for using oil Objectives.	Laboratory workers are advised to read the SDS and abide by the precautions documented. Laboratory worker shall successfully complete ? Chemical Safety Training? e-Learning module in the NTULearn portal or Workday@NTU. Wear the appropriate PPE; Lab coat, long pants & covered shoes. Use nitrile gloves when handling specimens. Apply BSL2 hygiene standards, washing hands at the end of the experiment.	1	3	3	NA
Physical	Others:Contact with heated surface	Burns/ scalds	Workers shall ensure that there is a 30 min interval between switching on and off the equipment. The lamp is in an enclosed housing which shall never be opened unless exchanging the lamp and unless the lamps has been off for at least 30 min to allow it to cool down. Use of the metal halide lamps shall not exceed 2000 hours. When the lamp usage approaches 2000 hours, only authorized personnel in SCELSE or service engineers from the equipment supplier are allowed to change the metal halide lamps.	2	1	2	NA